

Scalability & Future Plan

Date	2 July 2026
Team ID	6a3a555ae5895b2c9759029d
Project Name	EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current State

	JSON flat-file storage is not safe under concurrent writes	Risk of data loss/corruption under multi-user load	High	
S.No	Limitations			
	No user authentication or authorization	Cannot track individual learning history or restrict access	High	
	Single-process Uvicorn server, no caching layer	Limited throughput under heavy concurrent traffic	Medium	
1	Duplicate router-based and direct API endpoints	Increases maintenance overhead and API surface confusion	Medium	
2				
3				

Scalability Roadmap

	User Load	Single-process Uvicorn, JSON files	Deploy behind Gunicorn + multiple Uvicorn workers; add a reverse proxy / load balancer	
S.No	Scalability Area			
	Data Storage	JSON flat files (data/*.json)	Migrate to PostgreSQL or MongoDB while preserving the existing db_manager.py interface	
1	User Load	Performance	No caching; every request hits Gemini/T5	Add response caching for repeated topics; batch quiz generation requests
2	Data Storage	Security	No auth, open CORS	Add JWT-based authentication, restrict CORS origins, encrypt sensitive data at rest
3	Performance			
4	Security			

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2			
Phase 3			
Phase 4			

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